

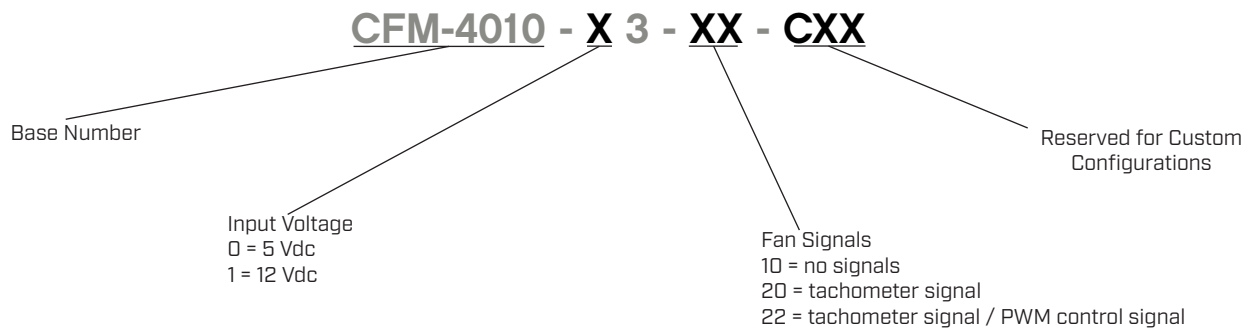
**SERIES:** CFM-40 | **DESCRIPTION:** DC AXIAL FAN**FEATURES**

- 40 x 40 mm frame
- high fan speed for greater air flow
- dual ball bearing construction
- auto restart protection standard on all models

**MODEL**

	input voltage		input current		input power	rated speed	air flow ¹	static pressure ²	noise
	rated [Vdc]	range [Vdc]	typ [A]	max [A]	max [W]	typ [RPM]	[CFM]	[inch H ₂ O]	max [dBA]
CFM-4010-03	5	4~5.75	0.16	0.24	1.2	8,300	10.0	0.29	37.0
CFM-4010-13	12	6~13.8	0.10	0.13	1.56	8,300	10.0	0.29	37.0

Notes: 1. At 0 inch H₂O static pressure.
2. At 0 CFM airflow.

PART NUMBER KEY

INPUT

parameter	conditions/description	min	typ	max	units
operating input voltage	5 Vdc input models	4	5	5.75	Vdc
	12 Vdc input models	6	12	13.8	Vdc
current	5 Vdc input models		0.16	0.24	A
	12 Vdc input models		0.10	0.13	A
power	5 Vdc input models		0.80	1.2	W
	12 Vdc input models		1.20	1.56	W
starting voltage	at 25°C				
	5 Vdc input models		4		Vdc
	12 Vdc input models		6		Vdc

PERFORMANCE

parameter	conditions/description	min	typ	max	units
rated speed	at 25°C, after 10 minutes	7,470	8,300	9,130	RPM
air flow	at 0 inch H ₂ O, see performance curves		10.00		CFM
static pressure	at 0 CFM, see performance curves		0.29		inch H ₂ O
noise	at 1 m		35.5	37.0	dBA

PROTECTIONS / SIGNALS¹

parameter	conditions/description	min	typ	max	units
auto restart protection	available on all models				
tachometer signal	available on “20” and “22” models				
PWM control signal	available on “22” models				

Notes: 1. See application notes for details.

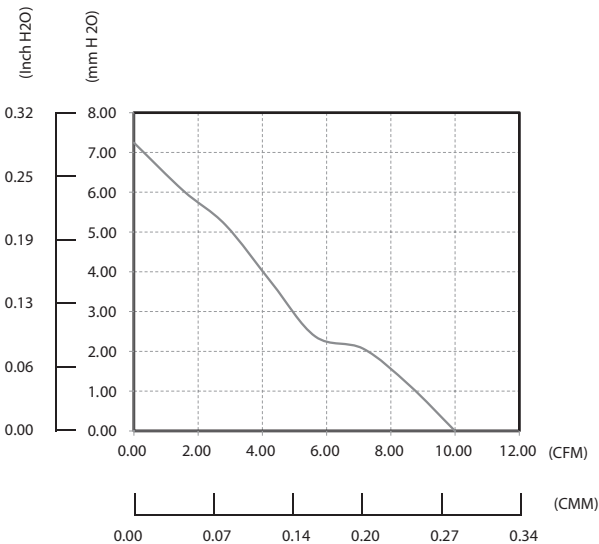
SAFETY & COMPLIANCE

parameter	conditions/description	min	typ	max	units
insulation resistance of frame	at 500 Vdc between frame and positive terminal	10			MΩ
dielectric strength	at 500 Vac, 60 Hz, 1 minute between frame and positive terminal			5	mA
safety approvals	UL/cUL 507, TUV [EN 62368-1]				
EMI/EMC	EN 55022:2010+AC:2011 Class B, EN 61000-3-2:2014, EN 61000-3-3:2013, EN 55024:2010				
life expectancy	at 45°C, 15~65% RH		70,000		hours
RoHS	yes				

ENVIRONMENTAL

parameter	conditions/description	min	typ	max	units
operating temperature		-10		70	°C
storage temperature		-40		70	°C
operating humidity	non-condensing	5		90	%
storage humidity	non-condensing	5		95	%

PERFORMANCE CURVES



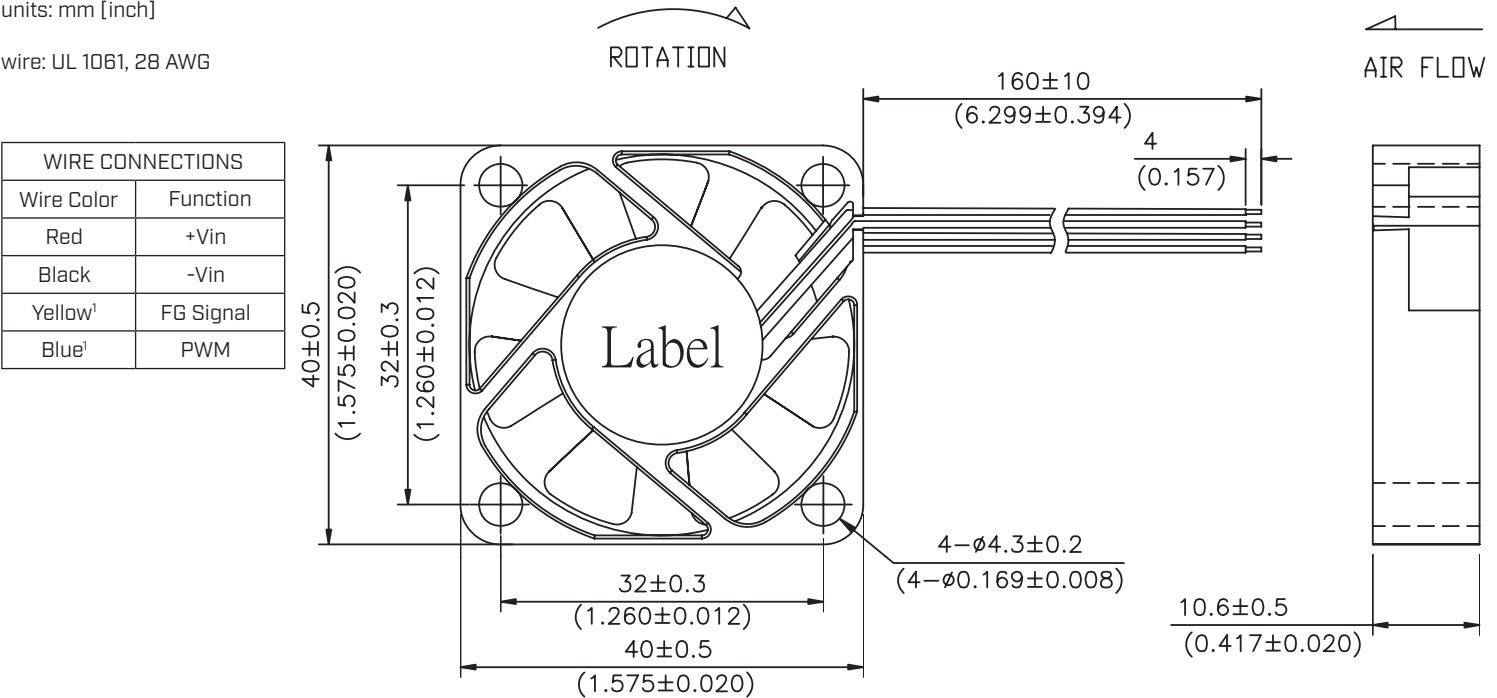
MECHANICAL

parameter	conditions/description	min	typ	max	units
motor	4 pole DC brushless				
bearing system	ball bearing				
direction of rotation	counter-clockwise viewed from front of fan blade				
dimensions	40 x 40 x 10.6				mm
material	PBT [UL94V-0]				
weight	5 Vdc input models		13.5		g
	12 Vdc input models		14.2		g

MECHANICAL DRAWING

units: mm [inch]

wire: UL 1061, 28 AWG

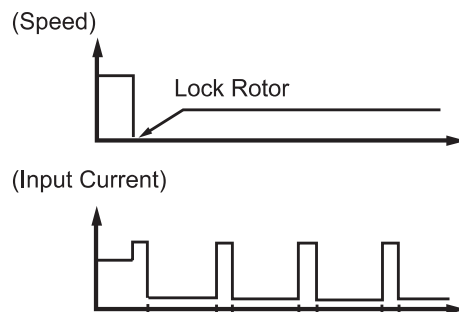


APPLICATION NOTES

Auto Restart Protection/Current Limit Protection

When the fan motor is locked, the device will cut off the drive current within two to six seconds and restart automatically after a few seconds. If the lock situation is continued, the device will work on a repeated cycle of cut-off and restart until the lock is released. [See Figure 1 below].

Figure 1 Current Limit Protection



Pulse Sensor/Tachometer Signal/FG

Pulse Sensor is for detecting the rotational speed of the fan motor. At locked rotor condition, the signal stops cycling and the output is fixed at VoH or VoL [See Figures 2~3 below].

Figure 2 Output Waveform

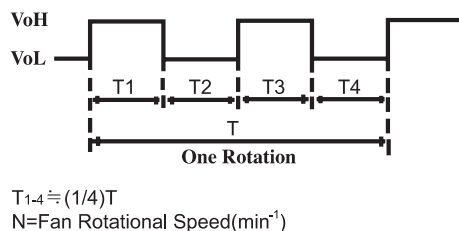
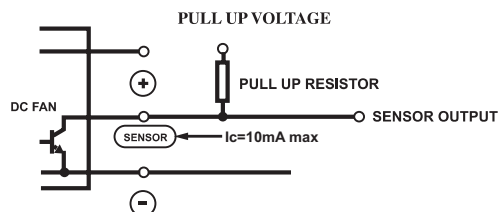


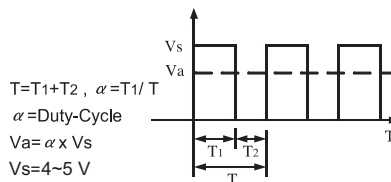
Figure 3 FG Signal Output Circuit: Open Collector



PMW Control Signal

A speed control lead can be provided that will accept a PWM signal from the customer circuit to vary the speed of the fan. The change in speed is linear by changing the Duty-Cycle of the PWM. Open collector type and pull-up voltage is changed by maximum operating voltage and sink current by consuming current. [See Figure 4 below].

Figure 4 Duty Cycle



REVISION HISTORY

rev.	description	date
1.0	initial release	08/15/2016
1.01	updated datasheet	07/27/2017
1.02	updated to be certified to EN 62368-1 safety standard	07/09/2019
1.03	brand update	02/07/2020
1.04	logo, datasheet style update	08/12/2022
1.05	CUI Devices rebranded to Same Sky	09/12/2024

The revision history provided is for informational purposes only and is believed to be accurate.



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